BOTTOM-ROW QUADRATIC REDUCTION AND SCALAR-SPAN OBSTRUCTION ON RANK-FOUR PARABOLIC ERDŐS–ULAM CONFIGURATIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. The Erdős–Ulam problem asks whether there exists a dense subset of $\mathbb{R}^2$ all of whose pairwise Euclidean distances are rational. In the Gaussian rank-four parabolic parameterization at scale $\lambda = 27$, we prove an exact quadratic reduction of the bottom distance row and a scalar-span rank-deficiency obstruction ruling out the rank-deficient non-star sub-branch.

1. Main Algebraic Reduction

Proposition 1.1. *On the Gaussian rank-four parabolic locus at parameter $\lambda = 27$, clearing the first three base-point rational-distance squares reduces the fourth distance condition to a binary quadratic norm equation whose scalar-span specialization has rank at most 2 and admits only collinear/star-degenerate rational solutions.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1994 in the domain of discrete and arithmetic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped algebraic obstruction* deciding the scope: quadratic reduction and rank-2 scalar-span obstruction on the Gaussian rank-4 parabolic lambda=27 locus.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- the unrestricted Erdos-Ulam rational distance problem.

References

[1] Primary problem list and source literature for Result 1994 (discrete and arithmetic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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